

CoRoT-1b

R-2205

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_250116 · created 2026-08-14T01:04:30+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460690.2581 +/- 0.0061 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.042
Transit depth (Rp/Rs)^2	2.05 +/- 1.19 [%]
Orbital Inclination (inc)	89.16 +/- 2.87
Airmass coefficient 1 (a1)	0.851 +/- 0.032
Airmass coefficient 2 (a2)	0.1 +/- 0.1
Scatter in the residuals of the lightcurve fit is	3.76 %
Best Comparison Star	#2 - [66, 416]
Optimal Method	PSF photometry
Transit Duration (day)	0.0495 +/- 0.007

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.143 ± 0.042 vs 0.1388 (deviation 0.1σ)
Duration fit vs published	0.0495 d vs 0.1046 d (deviation 7.71σ)
Mid-time O-C	-2188.6 min
Depth SNR	3.3
Residual scatter	3.76 %

Light curve

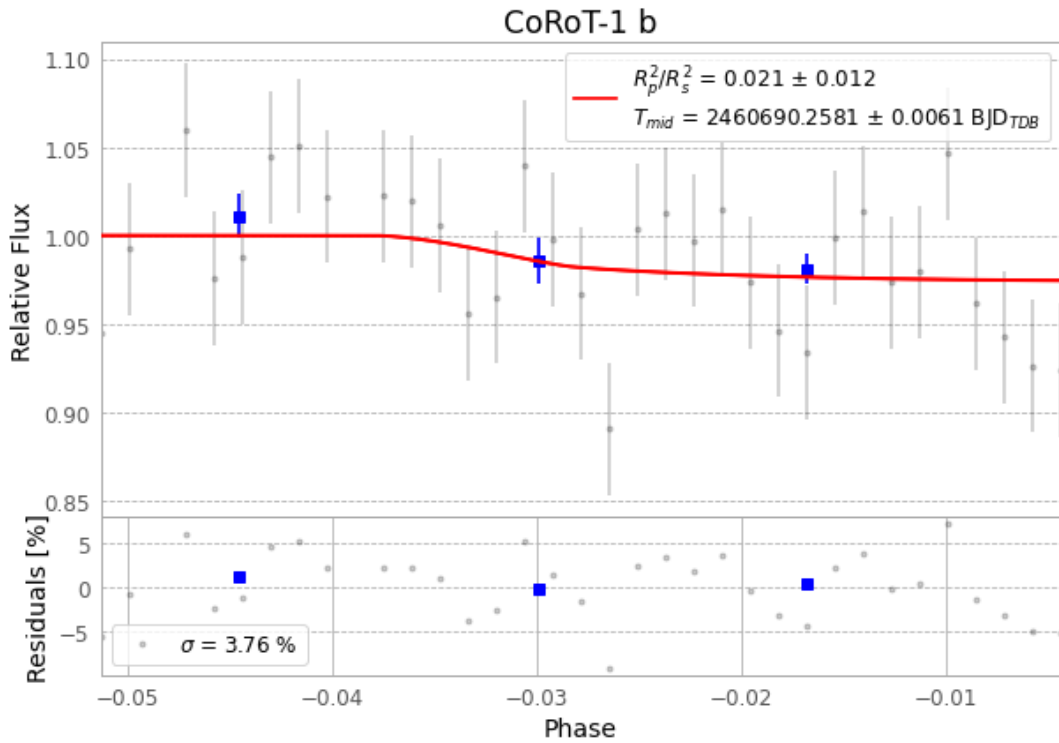


Figure 1 — FinalLightCurve_CoRoT-1 b_2025-01-15.png (SHA-256 b2f33f1b04ecb493...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [334, 209], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4734b40ee8f3d097...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

35 FITS frames (23 MB), CoRoT-1250116042415.FITS → CoRoT-1250116060616.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-250116.log (a737e90a5414...), FinalLightCurve_CoRoT-1 b_2025-01-15.png (b2f33f1b04ec...), FinalLightCurve_CoRoT-1 b_2025-01-15.csv (81643b04b079...)

Compute resources

Wall time 97.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 01:04Z.